

Simple Regression Part III – Assorted Comments

Question 1 (5 points)

Please access the Ames Housing Data. The dataset has too many variables to list here. We are **only** interested in the following regarding this problem:

- 2nd Flr SF (square footage of second floor)
- SalePrice (sale price of house in dollars)

Use JMP's Fit Model to fit the model $\text{SalePrice} = \alpha + \beta \times 2\text{ndFlrSF} + \varepsilon$. Include all the observations in the dataset in your analysis. According to the model fit, specify approximately how much more will a house sell for, with each one-square foot increase in the 2nd floor space (according to the model fit). Only one of the answers below is reasonable. Place an "X" in the box to the immediate left of your chosen answer.

Problem 1: Solution Choices	
X	\$50.23
	\$163945.30
	\$90.80
	\$1805.62

Question 2 (5 points)

This problem concerns the Ames Housing Data. The dataset has too many variables to list here. We are **only** interested in the following regarding this problem:

- 2nd Flr SF (square footage of second floor)
- SalePrice (sale price of house in dollars)

JMP's Fit Model was used to fit the model $\text{SalePrice} = \alpha + \beta \times 2\text{ndFlrSF} + \epsilon$. We used all the original data points, but we added one more that you don't have access to. Using information from the Analysis of Variance Table, below, determine the sample variance of the newly appended list of SalePrice values. Only one of the answers below is reasonable. Place an "X" in the box to the immediate left of your chosen answer.

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	1363215795524.67	1.363e+12	230.2219
Error	2929	17343526046090.9	5.9213e+9	Prob > F
C. Total	2930	18706741841615.5		<.0001*

Problem 2: Solution Choices	
	1,263,215,795,524.67
X	6,384,553,529.56
	5,334,163,235.38
	The videos are clear on this point: you can't determine the sample variance from the information given.

Question 3 (5 points)

This problem concerns the Ames Housing Data. The dataset has too many variables to list here. We are **only** interested in the following regarding this problem:

- 2nd Flr SF (square footage of second floor)
- SalePrice (sale price of house in dollars)

We used JMP's Fit Model to fit the model $\text{SalePrice} = \alpha + \beta \times 2\text{ndFlrSF} + \epsilon$. We used all the original data points, but added one more that you don't have access to. Using information from the Analysis of Variance Table, below, determine the coefficient of determination for the model fit to this newly appended data set. Only one of the answers below is reasonable. Place an "X" in the box to the immediate left of your chosen answer.

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	1363215795524.67	1.363e+12	230.2219
Error	2929	17343526046090.9	5.9213e+9	Prob > F
C. Total	2930	18706741841615.5		<.0001*

Problem 3: Solution Choices	
	0.728731
	230.2219
	0.007287
X	0.072873

Problem 4 (5 points)

This problem concerns the Ames Housing Data. The dataset has too many variables to list here. We are **only** interested in the following regarding this problem:

- 2nd Flr SF (square footage of second floor)
- SalePrice (sale price of house in dollars)

We used JMP's Fit Model to fit the model $\text{SalePrice} = \alpha + \beta \times \text{2ndFlrSF} + \epsilon$. We used all the original data points, but added one more that you don't have access to. Using information from the Analysis of Variance Table, below, determine the absolute value of the t statistic that would be used to test:

$H_0: \beta = 0$ vs. $H_A: \beta \neq 0$.

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	1363215795524.67	1.363e+12	230.2219
Error	2929	17343526046090.9	5.9213e+9	Prob > F
C. Total	2930	18706741841615.5		<.0001*

Only one of the answers below is reasonable. Place an "X" in the box to the immediate left of your chosen answer.

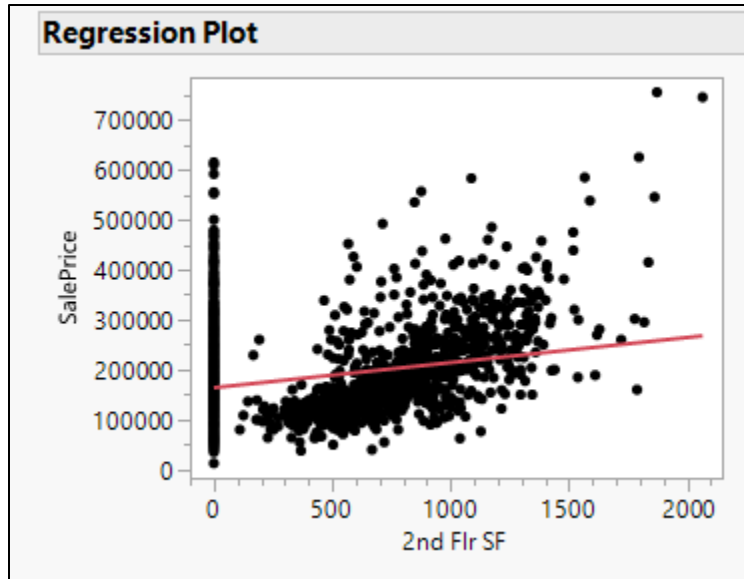
Problem 4: Solution Choices	
	10.38
	230.22
X	15.17
	90.79

Problem 5 (5 points)

This problem concerns the Ames Housing Data. The dataset has too many variables to list here. We are **only** interested in the following regarding this problem:

- 2nd Flr SF (square footage of second floor)
- SalePrice (sale price of house in dollars)

We used JMP's Fit Model to fit the model $\text{SalePrice} = \alpha + \beta \times 2\text{ndFlrSF} + \epsilon$. We used all the original data points and generated the scatterplot you see below. It appears that the dataset has a lot of zeros for 2ndFlrSF, probably corresponding exclusively to one-story houses. Specify what will happen to the estimated slope if the zeros are omitted before the model is fit. Only one of the answers below is reasonable. Place an "X" in the box to the immediate left of your chosen answer.



Problem 5: Solution Choices	
	It will necessarily become statistically meaningful.
X	It will increase.
	It will decrease.
	It will stay the same since a zero can have no effect.

Problem 6 (5 points)

Please access the Car Purchasing Data, with the following variables:

- Country
- Gender
- Age
- Annual Salary
- Credit Card Debt
- Net Worth

- Car Purchase Amount

Use JMP's Fit Model to fit the model $\text{CarPurchaseAmount} = \alpha + \beta \times \text{CreditCardDebt} + \epsilon$. Look at the JMP Parameter Estimates table on the output. Specify the hypothesis being tested corresponding to Credit Card Debt, and the outcome of that test. Only one of the answers below is reasonable. Place an "X" in the box to the immediate left of your chosen answer.

Problem 6: Solution Choices	
	$H_0: \beta = 0$ vs. $H_A: \beta \neq 0$, and the null is rejected.
	$H_0: \beta \neq 0$ vs. $H_A: \beta = 0$, and the null is not rejected.
	$H_0: \beta = 0.09$ vs. $H_A: \beta \neq 0.09$, and the null is rejected.
X	$H_0: \beta = 0$ vs. $H_A: \beta \neq 0$, and the null is not rejected.

Problem 7 (10 points)

Please access the Car Purchasing Data, with the following variables:

- Country
- Gender
- Age
- Annual Salary
- Credit Card Debt
- Net Worth
- Car Purchase Amount

Please answer each of the following questions. So that your answers remain focused, you are to include ONLY the output that is requested (else, your earned credit will be reduced).

- a. (5 points) Use JMP's Fit Model to fit the model $\text{CarPurchaseAmount} = \alpha + \beta \times \text{Age} + \epsilon$. Copy and paste the JMP Parameter Estimates table in the space below.

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	4742.5508	2193.378	2.16	0.0311*
Age	853.8259	46.75898	18.26	<.0001*

- b. (5 points) Refer to the Parameter Estimates table in part a. In three sentences or less, explain how you might use this information if you were managing a car sales team.

The model predicts that for each one-year increase in customer age, car purchase amount increases by approximately \$853. Because the relationship is statistically significant, a sales team could use age as one factor when estimating the price range of vehicles a customer may purchase.

Problem 8 (10 points)

Please access the Car Purchasing Data which has the following variables:

- Country
- Gender
- Age
- Annual Salary
- Credit Card Debt
- Net Worth
- Car Purchase Amount

Please answer each of the following questions. So that your answers remain focused, you are to include ONLY the output that is requested (else, your earned credit will be reduced).

- a. (5 points) Use JMP's Fit Model to fit $\text{CarPurchaseAmount} = \alpha + \beta \times \text{CreditCardDebt} + \epsilon$. Copy and paste the JMP Parameter Estimates table in the space below.

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	43353.018	1413.488	30.67	<.0001*
Credit Card Debt	0.0891771	0.1383	0.64	0.5193

- b. (5 points) Refer to the Parameter Estimates table in part a. In three sentences or less, explain how you might use this information if you were managing a car sales team.

The model predicts that for every one dollar increase in credit card debt, car purchase amount increases by about 9 cents. However, this relationship is not statistically significant, so a sales team should not rely on credit card debt as a useful predictor of how much a person will spend on a car.